

MORPHOMETRIC EVOLUTION OF COMPOSITE VOLCANOES

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Abstract. Statistical relations have been determined between geometry, volume, slope, and age for 26 circum-Pacific composite (strato) volcanoes. General trends in eruption characteristics, repose periods, flow lengths, and petrology are also documented. Few examples of the earliest stages of composite volcano activity are known, perhaps because these small volcanoes are indistinguishable from cinder cones. If cinder cones evolve into composite volcanoes a fundamental change in morphometry, eruption style, and petrology occurs at a basal diameter of 2 km.

Composite volcanoes (stratovolcanoes), composed of layered deposits of pyroclastics and lavas, are the characteristic volcanic landform at subducting plate margins, and are the most abundant type of large volcano on the Earth's surface. Composite volcano morphology results from repeated eruptions of pyroclastics and relatively short lava flows from a central vent. By comparison, pyroclastics are insignificant and lava flows tend to be much longer (10-100 km) for shield volcanoes. Most composite cones are formed of andesites or basaltic andesites, but some are composed of basalts (Fuji, Fuego, Izalco); thus petrology may be less important in determining the morphology of composite cones than eruption style, as is true for shield volcanoes (Wood, 1977a). In this study, observations of eruption characteristics and chemical/petrological trends for a number of composite cones were synthesized with newly determined measurements of cone morphology to document statistically the evolution of composite volcanoes and to examine their relation to cinder cones. Additionally, such quantitative descriptions of terrestrial volcanoes provide basic data for comparisons with volcanic structures on Mars and the Moon.

Morphometry

Ideally, to determine a volcano's original or constructional geometry, only uneroded, isolated cones should be considered, yet the eruptive process itself erodes flanks and modifies vents. Practically, the availability of topographic and bathymetric maps determines which volcanoes can be measured, and only those with obvious calderas or other major modifications are excluded. Preliminary measurements for 26 composite cones (most historically active) from the circum-Pacific belt document systematic relations in cone morphometry (Fig. 1) and, since the cones are at different stages of growth, evolutionary development. The cones have the following ranges of dimensions (km): basal width (Wco) = 0.6-22; summit crater width (Wcr) = 0.03-0.7; crater depth (Dcr) = 0.03-0.45; cone height (Hco) = 0.2-3. For these data, least squares analyses reveal consistent linear variations that suggest geometrically uniform cone growth:

$$Hco = 0.122Wco + 0.450 \quad (n=17; r=.95) \quad (1)$$

$$Wcr = 0.027Wco + 0.048 \quad (n=14; r=.91) \quad (2)$$

$$Wcr = 0.222Hco - 0.054 \quad (n=18; r=.92) \quad (3)$$

There is no significant ($r=.62$) relation between crater depth and diameter. Volcano volume (V) can be estimated using the equation for a cone ($V = (\pi/3) (Wco/2)^2 (Hco)$) or, if either Hco or Wco is unknown:

$$V = (Wco)^2 (0.032 Wco + 0.117) \quad (4)$$

$$V = Hco (17.6 (Hco)^2 - 15.9 Hco + 3.58) \quad (5)$$

The sampled cones range in volume from 0.17 - 395 km³. Equation (1) may be used to estimate original heights (and hence volumes of missing summit material) of composite cones that have collapsed to form calderas. For example, the basal diameter of Mt. Somma, the ancestral cone of Vesuvius, is ~15 km, implying a pre-collapse height of 2.3 km, in good

accord with Imbo's (1965) estimate of 2.5 km based on extrapolation of slopes.

Average slopes ($S = \arctan [2Hco / (Wco - Wcr)]$) for composite volcanoes decrease exponentially (Fig. 2) from ~33° for cone diameters of 2 km, to ~15° for cones 22 km wide:

$$S = 41.2 Wco^{-0.33} \pm 3^\circ \quad (6)$$

This relation supports a previous assumption (Francis and Abbott, 1973) that cone height and slope are inversely related.

Eruption Characteristics

Eruptions of composite volcanoes are variable in style, duration and frequency. Many composite cones exhibit nearly continuous mild strombolian activity, occasionally interrupted by paroxysmal eruptions of greater violence and volume (Izalco, Fuego, Stromboli). Average mild strombolian explosions at Stromboli produce about 0.065-0.005 m³ of ejecta once every ~11 minutes (Chouet *et al.*, 1974). At this rate ~10¹¹ explosions (1.7 x 10⁶ yrs) would be necessary to construct Izalco, a cone that initially erupted in 1770 AD. Thus mild strombolian activity is insignificant in cone construction. Intense explosive activity ejecting both juvenile and non-juvenile material (vulcanian eruptions) often results in small pyroclastic flows (Fuego, Cerro Negro, Ngauruhoe) with volumes of 10⁶-10⁷ m³ (Rose *et al.*, 1973). More important changes in cone morphology occur during cataclysmic peleeen eruptions (Pelee, Santa Maria) that produce rapidly moving nuee ardentes often associated with formation of viscous domes (Roobol and Smith, 1975). The least frequent, but most extreme, style of activity is plinian eruption with caldera collapse and voluminous (km³) pumice deposition (Krakatoa, Coseguinia). Although caldera formation often appears to be the final stage of volcano evolution, in numerous cases activity resumes with a new cone rebuilding the destroyed summit (Vesuvius, Fuji).

Growth Rates

Only a few composite volcanoes have formed within recent centuries and generally few details are known for these eruptions. Since Izalco was born in 1770 a cone ~0.76 km³ has been constructed, for an average eruption rate of 0.41 km³/100 yrs, although during periods of intense activity (e.g., 1951-56) the rate is 10 times greater. In the first 121 years of its lifetime Cerro Negro has averaged 0.15 km³/100 yrs. For comparison, Kilauea, a basaltic shield volcano, has averaged 1.2 km³/100 yrs for the 51 years ending in 1969. In a preliminary examination of growth rates over geologic periods, estimated ages (A) and basal widths (Wco) of 9 composite cones were compared (Fig. 3). For cones between 121 and 5 million years old:

$$Wco = 0.63A^{0.18} \pm 0.65 \text{ km} \quad (7)$$

with younger cones growing faster and older ones slower.

Repose Periods

For 20 Central American composite cones with long records listed by Mooser *et al.* (1958) repose periods are most frequently 1 year or less; this statistic is biased, however, by separate entries for frequent mild eruptions at Fuego and Izalco which really represent persistent activity. Excluding repose periods < 1 yr yields a median value of 7 yrs. Longer repose periods are common (the longest being 302 yrs for Santa Ana); a number of volcanoes regularly active in the 17th century (Chichigalpa, Santa Clara, El Viego) have been in repose nearly 400 yrs, while other

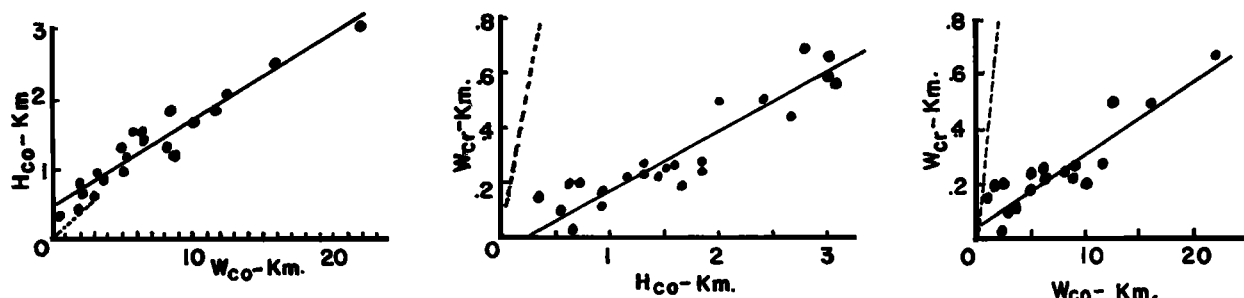


Fig. 1. Relations between basal diameter (W_{co}), summit crater diameter (W_{cr}), and cone height (H_{co}) for composite volcanoes (data points and solid line least squares fit) and cinder cones (dashed lines; Porter, 1972).

cones with no previous historic activity have erupted unexpectedly (Santa Maria, Ceboruca, Acatenango). Suggestions that repose periods increase with volcano age (e.g., Rose *et al.*, 1977) are reasonable, but no systematic relation has been found between repose period and cone diameter ($\sim$ age).

Flow Lengths

During each phase of activity lava flows may erupt from the flanks or the summits of composite cones. These flows are usually short (average length of 34 recent flows at Izalco, Nguarohoe, Ceboruca, Momotombo and Cerro Negro is 1.95 km) and often thick. These properties have led to assumptions that such flows are viscous dacites, however, petrological examination shows that many short flows thought to be dacites are actually andesitic lavas (e.g., Toliman flows, Williams, 1960) whose high viscosity must reflect physical rather than chemical controls (e.g., effusion rate rather than SiO_2 content). One possibility is that persistent venting of gases during mild strombolian activity at summit craters devolatilizes magma resulting in flank eruptions of relatively viscous lavas. A similar degassing frequently initiates more violent eruptions which are often characterized by summit explosions preceding lateral lava effusions.

Petrological/Chemical Trends

Only a few volcanoes have been sampled at close enough intervals to document systematic changes in composition, although gross changes (andesitic cone with dacite summit plug) are well known. Generally, tephra and lavas become more silicic as a volcano grows – this is well documented for Santa Maria and the Myoko cones. At Santa Maria SiO_2 increases up a 240 m section from 52% to 57% at the summit and there are associated systematic increases in K_2O , Rb and Zr, and decreases in MgO and CaO (Rose *et al.*, 1977). These variations were interpreted in terms of increasing crystal fractionation due to cone growth and increasingly long repose periods. Similar chemical changes are observed in the Myoko volcanoes (Hayatsu, 1977), with petrology of the eruption products changing with eruption stage: During cone growth pyroxene-olivine basalts and pyroxene andesites predominate, but following caldera col-

lapse hornblende-pyroxene andesites are most common. Late stage eruptions of hornblende-andesites and dacite build an inner cone or dome on the caldera floor. The rate of SiO_2 enrichment at Izalco and Cerro Negro is $\sim 2.2\% \Delta SiO_2/100$ yrs, but at Santa Maria it is only $0.03\%/100$ yrs. The differences between these two rates may reflect the periods represented (~ 200 yrs vs. ~ 13000 yrs, respectively) or the evolutionary stage of the volcanoes, but faster crystal settling in the more mafic magma of Cerro Negro and Izalco (49-54% SiO_2 vs. 52-57% for Santa Maria) may also be important. Major evolutionary changes within composite cone magma chambers are indicated by (1) the common occurrence of hornblende in late stage eruptions (Santa Maria, Shastina, Myoko), (2) late stage satellitic eruptions of basaltic magmas while acidic eruptions continue at the summit (Shasta, Rainier; Condie and Swenson, 1973), and (3) the common construction of composite cones on basaltic shield volcanoes (Thielson, Makalia).

Comparison of Composite and Cinder Cones

Small composite cones are rarely observed and Crowe (1977) proposed that cinder cones, strengthened by dike and sill emplacement, evolve into large composite volcanoes. Mt. McLoughlin, a strongly dissected composite volcano in the Cascades, is a possible example for this process for it encapsulates a large cinder cone (Harris, 1976). Although the two types of cones may be related there are a number of significant differences between them. Cinder cones are considerably smaller (average basal diameter = 0.9 km vs. 8 km for composite cones) and have different dimensional relations. The principal difference is in crater diameters: for a cinder cone with a basal diameter of 2 km, the summit crater is 0.8 km wide, whereas a composite cone of the same basal size has a crater only 0.1 km in diameter. The wide craters of cinder cones are principally due to ballistic deposition of ash at low enough velocities for ash to accumulate around the vent (Chouet *et al.*, 1974). Cinder cones are further enlarged by slumping and rearing (Wood, 1977b). Frequently ash from composite volcano eruptions is lofted so high that it does not contribute to the crater ramparts, which are built by lava flows and less energetic ash eruptions. These small eruption vents are considerably enlarged by ream-

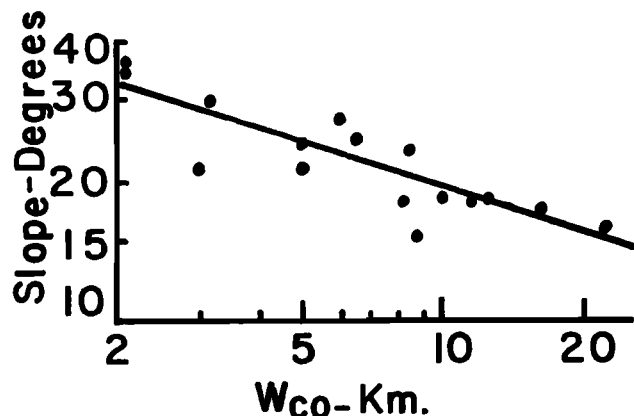


Fig. 2. Variation in average slope and basal diameter (W_{co}) for composite volcanoes.

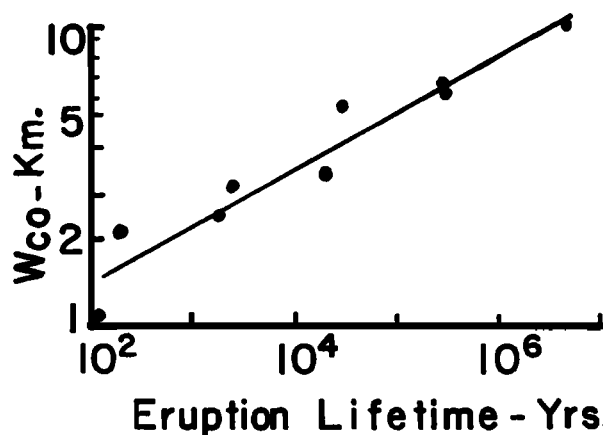


Fig. 3. Growth of composite volcano basal diameter (W_{co}) with time.

ing during violent eruptions (Vesuvius, 1906) or wholesale collapse following catastrophic eruption (Coseguinia, 1835). Commonly, cinder cones are basaltic, whereas composite volcanoes are typically andesitic. Complete eruptive histories for cinder cones range from a few days (Monte Nuovo) to ~ 10 yrs (Paricutin), whereas composite volcanoes have lifetimes of 10^4 - 10^6 yrs. Cinder cones typically result from a single eruptive episode with later volcanism producing new cones at separation distances of a few km (McGetchin and Settle, 1975). Composite cones are built by thousands of eruptions from the same vent with prolonged repose periods common. Strombolian eruptions dominate cinder cone growth, but all styles of eruptions occur in composite cone activity, and frequently a succession of volcanoes form with only slight vent offsets (Vesuvius). These differences indicate that if cinder cones and composite volcanoes are related, fundamental changes in morphometry, eruption style, and petrology occur when the basal diameter increases beyond ~ 2 km, the approximate diameters of the largest cinder cones.

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References

- Chouet, B., N. Hamisevicz, and T.R. McGetchin, Photoballistics of volcanic jet activity at Stromboli, Italy, *J. Geophys. Res.*, 79, 4961, 1974.
- Condie, K. and D. Swenson, Compositional variation in three Cascade stratovolcanoes: Jefferson, Rainier, and Shasta, *Bull. Volc.*, 37, 205, 1973.
- Crowe, B., Composite cone growth modeled after Broken Top, a dissected High Cascade volcano, *Geol. Soc. of Amer. Abstr. with Programs*, 9, 940, 1977.
- Francis, P.W. and B.M. Abbott, Sizes of conical volcanoes, *Nature*, 244, 22, 1973.
- Harris, S.L., *Fire and Ice*, p. 78, Pacific Search Books, Seattle, Washington, 1976.
- Hayatsu, K., Geologic study on the Myoko volcanoes, Central Japan, *Mem. Fac. Sci.*, Kyoto Univ. (Ser. Geol. and Min.), 43, 1, 1977.
- Imbo, G., *Italy, Catalog Act. Volc. World*, Pt. 18, Intern. Assn. Volc., Naples, Italy, 72, 1965.
- McGetchin, T.R. and M. Settle, Cinder cone separation distances; implications for the depth of formation of gabbroic xenoliths, *EOS*, 56, 1070, 1975.
- Mooser, F., H. Meyer-Abich, and A.R. McBirney, *Central America, Catalog Act. Volc. World*, Pt. 6, Intern. Assn. Volc., Naples, Italy, 146, 1958.
- Porter, S., Distribution, morphology, and size frequency of cinder cones on Mauna Kea Volcano, Hawaii, *Geol. Soc. Amer. Bull.*, 83, 3607, 1972.
- Roobol, M.J. and A.L. Smith, A comparison of the recent eruptions of Mt. Pelee, Martinique and Soufriere, St. Vincent, *Bull. Volc.*, 39, 214, 1975.
- Rose, W.I., S. Bonis, R.E. Stoiber, M. Keller, and T. Bickford, Studies of volcanic ash from two recent Central American eruptions, *Bull. Volc.*, 37, 338, 1973.
- Rose, W.I., N.K. Grant, G.A. Hahn, I.M. Lange, J.L. Powell, J. Easter, and J.M. Degraff, The evolution of Santa Maria volcano, Guatemala, *J. Geol.*, 85, 63, 1977.
- Williams, H., Volcanic history of the Guatemalan highlands, *Univ. Cal. Pub. Geol. Sci.*, 38, 1, 1960.
- Wood, C.A., Non-basaltic shield volcanoes, *Abst. Planet. Geol. Field Conf. Snake River Plain, Idaho*, 35, NASA TM-78, 436, 1977a.
- Wood, C.A., The growth of cinder cones, paper presented at the IASPEI-IAVCEI Assembly, Durham, 1977b.

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